

2.2 Coastal landforms

Coastal landforms of erosion

The geology of an area helps determine the shape of the coastline and its landforms. This is due to the resistance (hardness) of the rock. Harder rock, such as chalk, erodes at a much slower rate than softer rock, such as clay. A **concordant coastline** has different rock types along its length. Alternating bands of hard and soft rock run parallel to the shoreline. This results in even rates of erosion along the coast. A **discordant coastline** has different rock types exposed to the sea. This can produce coastlines with **bays** and **headlands** as the rock types erode at different rates. Headlands are formed from harder rock which is more resistant to erosion. Bays are formed when softer rock is eroded at a faster rate than the adjacent headlands.

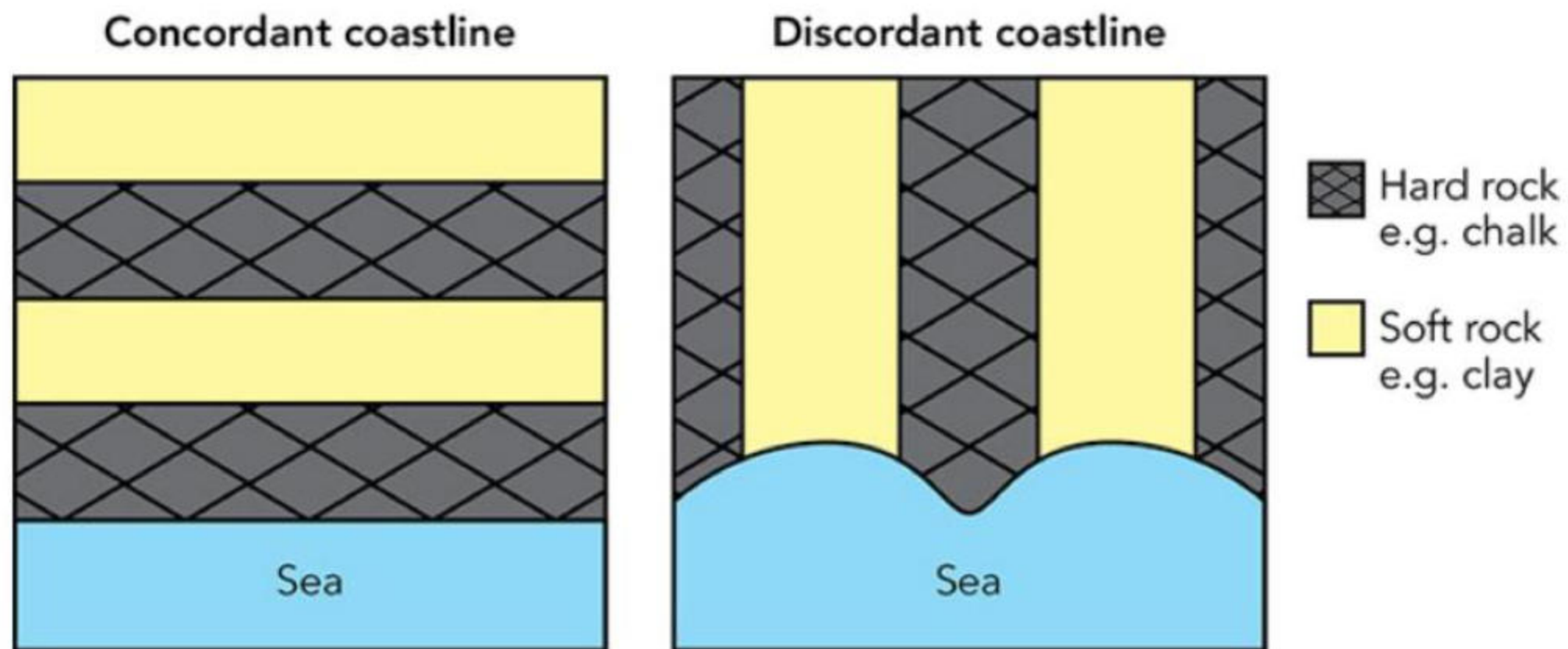


Figure 2.6: Concordant and discordant coastlines

The erosion processes of hydraulic action, corrasion, and corrosion erode headlands to form a **cliff**. As a result of wave action at the base of a cliff, a wave-cut notch can develop over time between the points at which low tide and high tide reach. This results in undercutting at the base of the cliff, eventually leading the cliff to collapse, as it is no longer supported. As the cliff retreats, a **wave-cut platform** is left behind.

Cracks within a cliff at a headland are widened over time by processes of erosion to form a **cave**. This cave grows deeper due to erosion and eventually breaks through to the other side of the headland. This leaves behind an **arch**. If the roof of this arch collapses, a **stack** is formed. Erosion can then wear a stack down over time to form a **stump**.

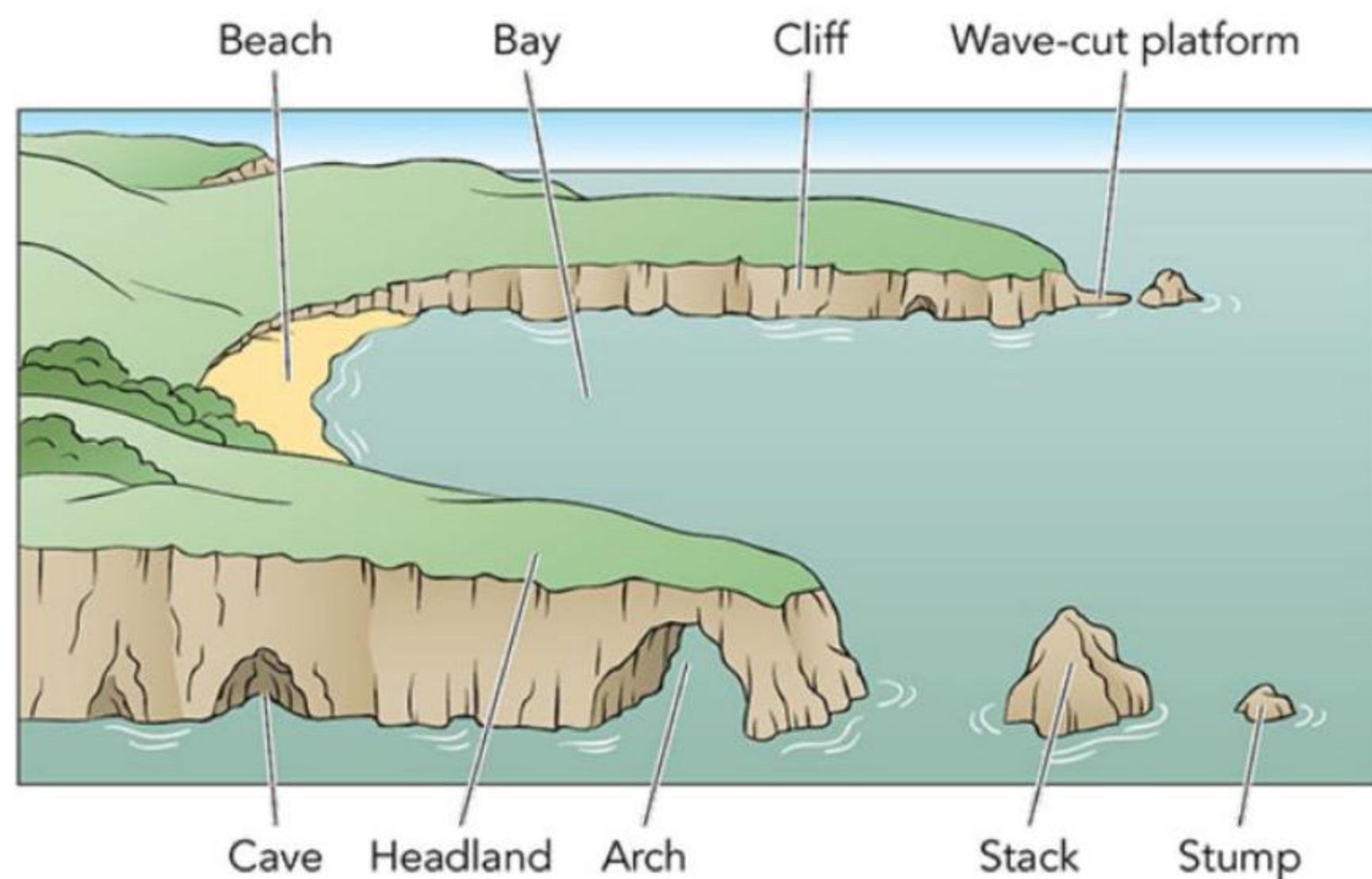


Figure 2.7: Coastal landforms



Source **A**: A coastal landform at Legzira Beach, Morocco

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Source **B**: The Twelve Apostles, Australia

TASK 1: Study Sources **A** **B**

- a Identify the coastal landforms you can see.
- b **Using diagrams**, explain how each landform was formed.

REFLECTION

When explaining how a coastal landform is created, does drawing diagrams help you understand the process more easily? Why is this?
What makes an effective diagram in geography?

Coastal landforms of deposition

Depositional landforms occur where there has been a build-up of sediment at a point along the coastline. A **beach** is a common landform of deposition and occurs in a sheltered area, such as a bay, where waves deposit sand or pebbles. Constructive waves help build up a beach, as they have a greater swash than backwash.

A **spit** forms where there is a change in direction of the coastline, such as at a headland or before an estuary (the mouth of a river). This is where waves deposit sand or pebbles which have been transported by longshore drift. Over time, this accumulation of sediment extends into the water. This forms a narrow, elongated landform known as a spit (see Figure 2.8). If the spit continues to grow and is influenced by changes in wave patterns or wind direction, it may curve at its outer end. This forms a hook. Spits are dynamic features, and their shape and size can change over time due to the ongoing processes of erosion, deposition, and the movement of coastal sediments. They play important roles in coastal ecosystems, providing a habitat for various plants and animals. Often a marsh develops behind a spit. If a spit continues to grow across a bay and connects two headlands together, it forms a coastal **bar**.

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TASK 2: Study Source C

- a If the two headlands were connected by the spit, identify what landform would be created.
- b **Produce a field sketch** of the landform in Source C and label its key features.



Source C: Farewell Spit, New Zealand

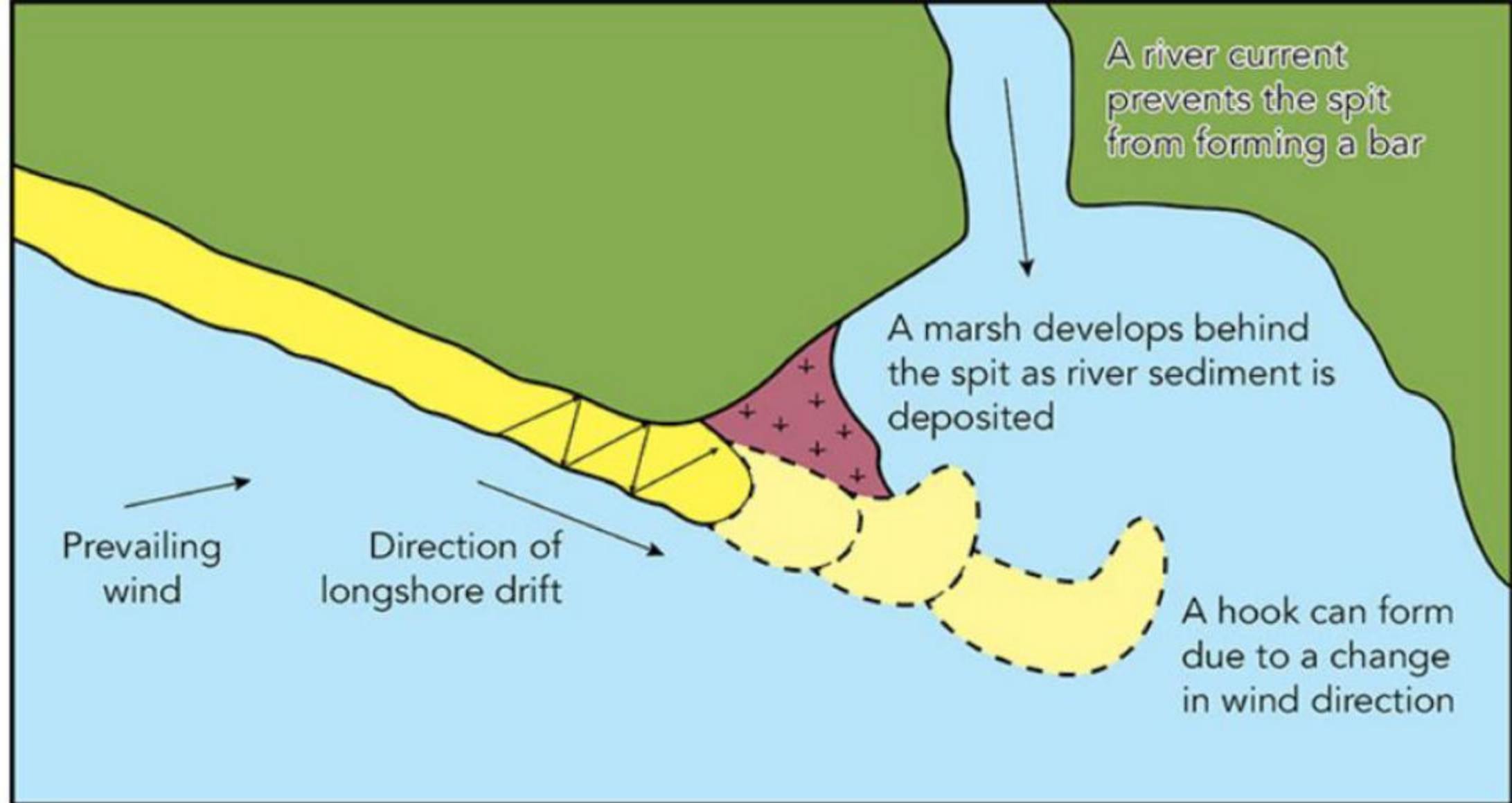


Figure 2.8: The formation of a spit

Coastal **sand dunes** are formed from loose sand that winds blow onshore. Vegetation and debris on the beach slow the wind down, causing the sand to be deposited. As the sand accumulates, it forms small mounds. Over time, these mounds grow into larger dunes as more sand is deposited. Vegetation then grows which stabilises the dunes. As more dunes form, the **ecosystem** develops.



Figure 2.9: A coastal bar at Amanohashidate in Japan



Figure 2.10: Coastal sand dunes in Jutland, Denmark

TASK 3: Study Source **D**

- a** Study the map extract in Source **D** and identify the depositional landform shown.
- b** Measure the length of the landform across Slapton Ley.

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Source **D:** Map extract from the UK coastline. Map scale 1:25 000

TASK 4

- a** Use an atlas or an online map to identify the landforms found at the following coordinates:
- i** $43^{\circ}15'22.9''\text{N } 16^{\circ}38'02.6''\text{E}$
 - ii** $58^{\circ}53'09.9''\text{N } 3^{\circ}25'52.2''\text{W}$ (you may need to look in 3D online)
 - iii** $19^{\circ}17'40.3''\text{N } 155^{\circ}05'52.5''\text{W}$ (you may need to look in 3D online)
- b** Using an atlas or online map, identify as many examples as possible of the coastal landforms in this topic along a coastline near you.